

Java Institute For Advanced Technology

Web Programming II, SRS Documentation

HF2L 04/AS/02

Name :	Nimuthu Methmadu Paranawithana
Instructor :	Anjana
Course :	Web Programming II
NIC :	200418100551
Branch :	Colombo Branch





Software Requirements Specification

Version 1.0
February 17 2026

Hercey Organic Online Market System

Prepared by Nimuthu Paranawithana

Submitted in partial fulfillment Of the requirements of
CS 310 Software Engineering

Table of Contents

1. List of figures …… 4
2. Introduction …… 5
   - 2.1. Purpose
   - 2.2. Scope of project
   - 2.3. Glossary
   - 2.4. References …….. 6
   - 2.5. Overview of Document
3. Overall Description
   - 3.1. System environment
4. Users and Characteristics ……. 7
5. External Interface Requirements
6. User Interface
   - 6.1. Home
7. External Interface Requirements …… 16
   - 7.1. User Interfaces
   - 7.2. Server hardware ……. 18
   - 7.3. Software Interfaces
8. Functional Requirements ……. 20
9. Behavior Requirements ……. 21
10. Performance Requirements
11. Safety and Security Requirements ……. 22

1. List of figures

System environment Diagram	**1**
System environment	**6**
UML use case	**21**

2. Introduction

2.1. Purpose

The purpose of this documentation is to present a detailed script of the Hercey Online Market Web application. This will explain each and every feature of the web application and its purpose and its effect on the web application, the conditions which it must operate under. This document is intended for both the stakeholders and the developers of the system.

2.2. Scope of Project

This web application will be a central market system for all herbal and tea based products. The goal is to connect all local tea and herbal producers island wide with quality assurance and monitoring. This system will extend the reach of local stores to overseas and island wide local buyers. The system will maximize the quality of the product since every product or batch is audited by an admin for presentation quality and product itself. Every product and seller goes through a verification process by admins.All users are verified by email. The system's primary focus is tea and other herbal products of similar but lower priority level.

2.3. Glossary

Term	Definition
Database	Collection of all the information monitored by this system.
Software Requirements Specification	A document that completely describes all of the functions of a proposed system and the constraints under which it must operate. For example, this document.
Stakeholder	Any person with an interest in the project who is not a developer.
User	Buyer or seller.
Web application	social media platforms, content management systems, and any application where content changes based on user input or database updates
Hercey	The name incorporated to the online web store application
Session cart	A cart that works in the session
DB cart	A cart that save items in data base (permanent)

2.4. References

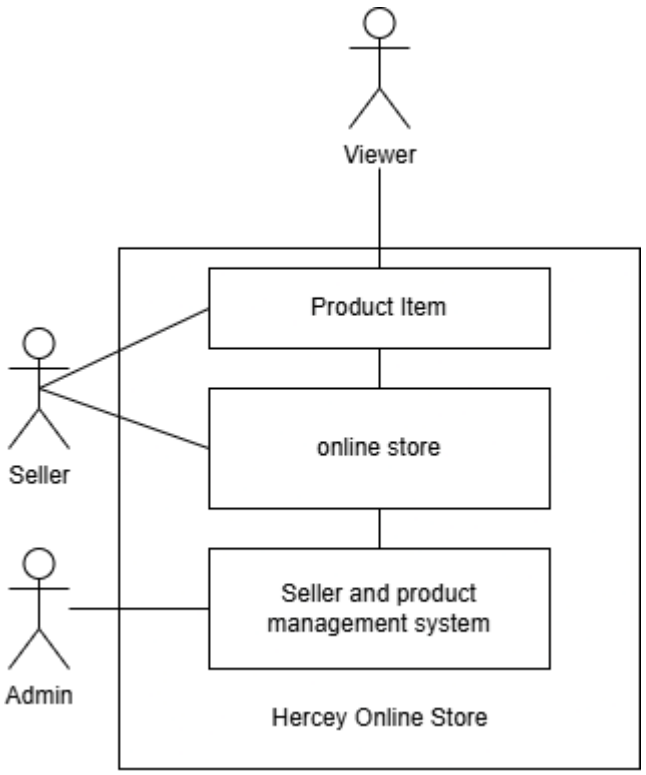
IEEE. IEEE Std 830-1998 IEEE Recommended Practice for Software Requirements Specifications. IEEE Computer Society, 1998.

2.5. Overview of Document

The next chapter, the Overall Description section, of this document gives an overview of the functionality of the product. It describes the informal requirements and is used to establish a context for the technical requirements specification in the next chapter

3. Overall Description

3.1. System environment



Hercey has 3 active actors and one central system. Seller, Buyer, and the Admin. Any seller or product goes through the management system to get approval for the admins and the admins can access the entire system directly. Only approved products will be shown to the buyers.

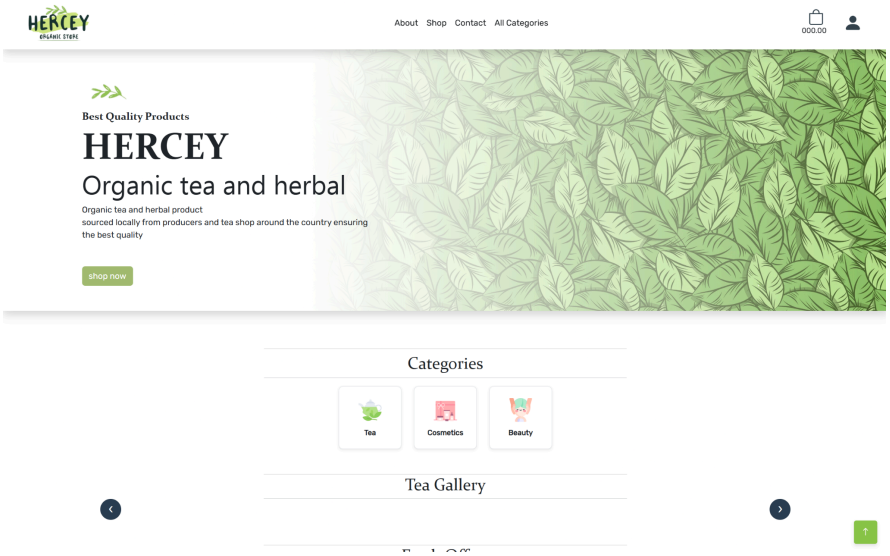
4. Users and Characteristics

The main target audience is overseas sellers/ business owners and local customers. The web application uses intuitive icons and legacy naming conventions in mix with modern designs contrasting the most frequently used features like B2B functions and purchase options. We assume that people with common knowledge with computing and web navigation will be interacting with this web application.
We plan on incorporating live AI chat support to the web application for average user questions and live customer support for sensitive and complicated problems.

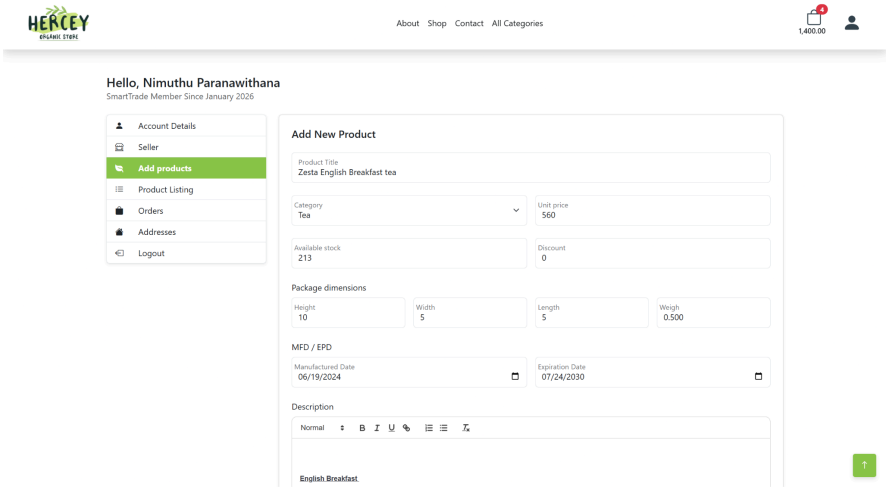
5. External Interface Requirements

5.1. User Interface

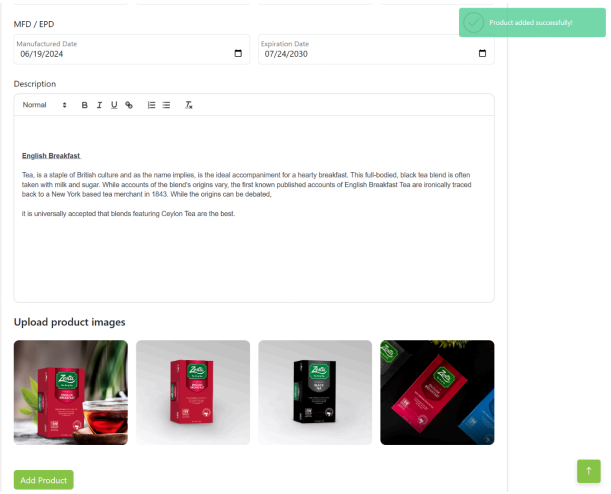
5.1.1. Home



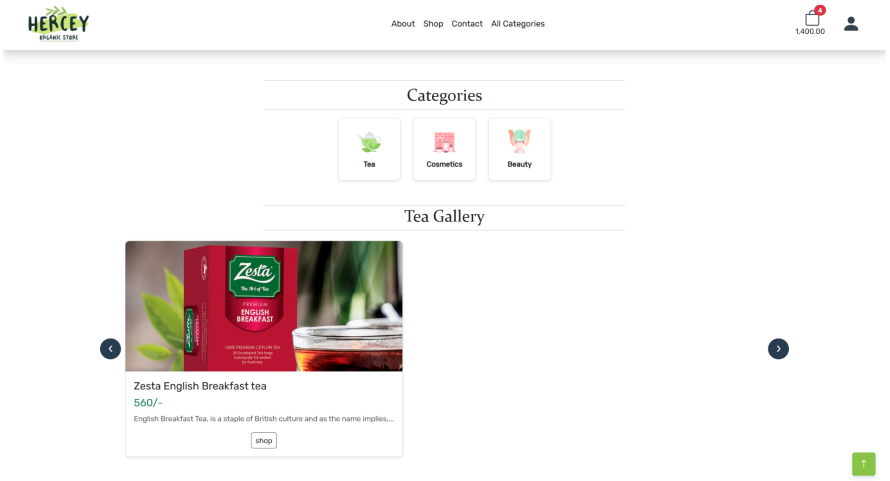
The home interface has quick links to About, Shop, Contacts, and All product categories and a profile menu and a cart icon to navigate to cart in the header. Right after the hero section we have a categories section showcasing all the categories in the shop and a section for a tea show case. Which is empty in the picture since there are no items registered in the shop.



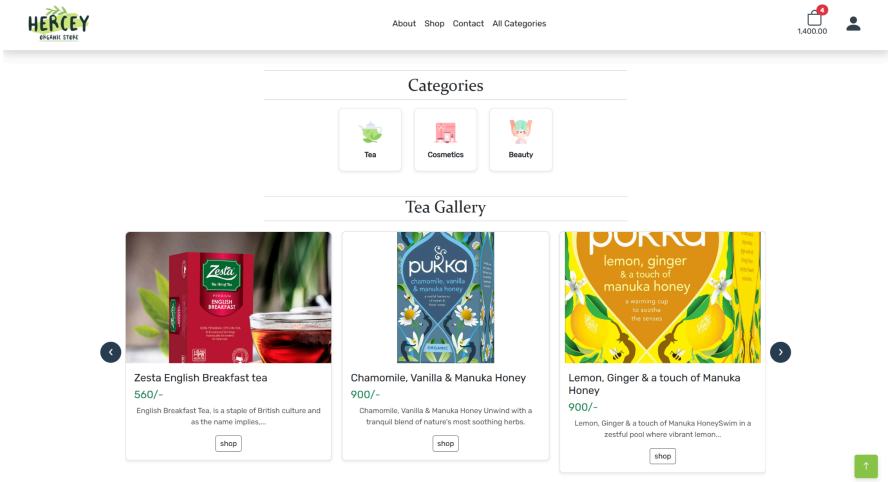
Here we can see the add product section for a seller. Adding a product requires a title, category, price, stock, discount (Optional), dimensions, MFD EXD, description and up to 4 pictures, 1st being the primary picture.



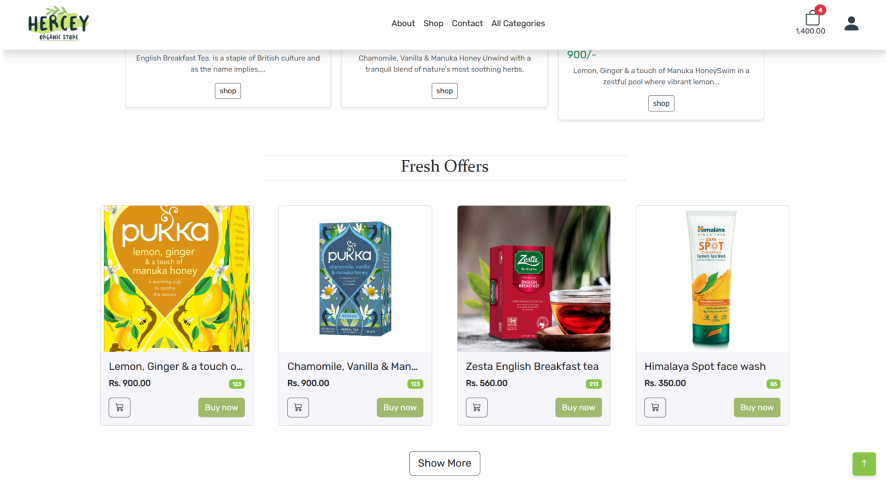
After adding a Tea product and the approval from the admin the product will appear in the gallery. Categories are also dynamically loaded according to admins.



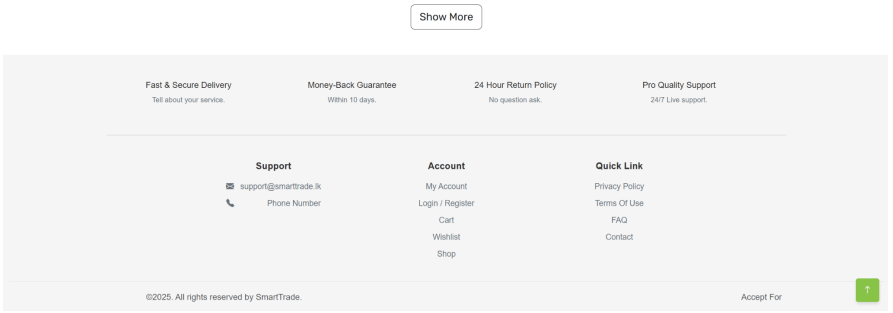
After products are loaded the tea gallery will work as a carousel



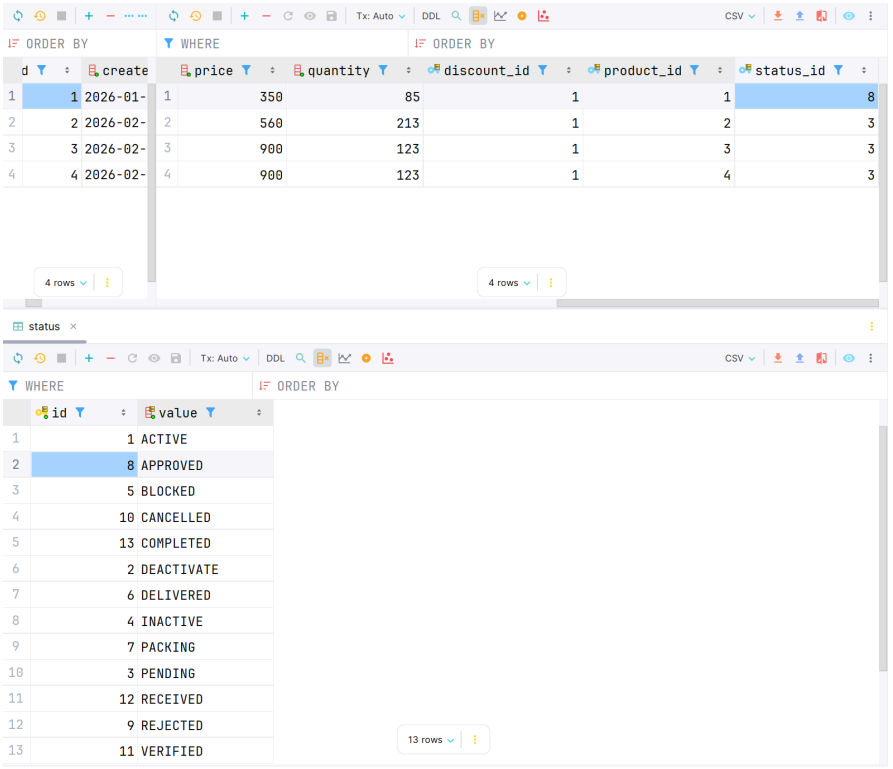
After the gallery user can see the 12 newest additions to the store according to the added time.



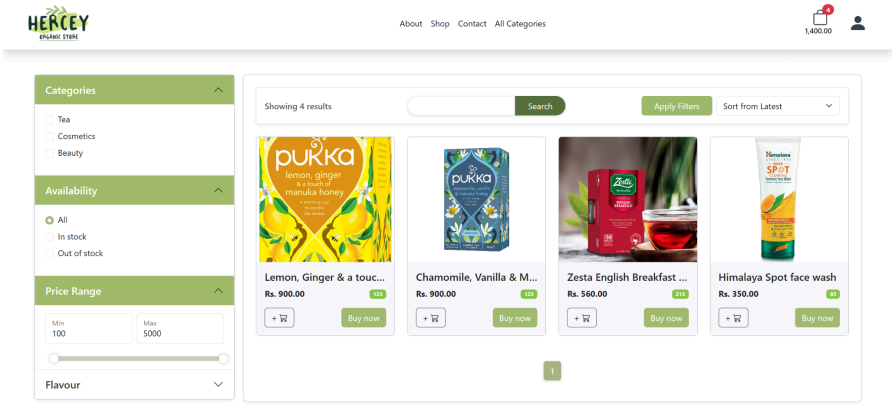
And then a footer with useful links, in almost every page



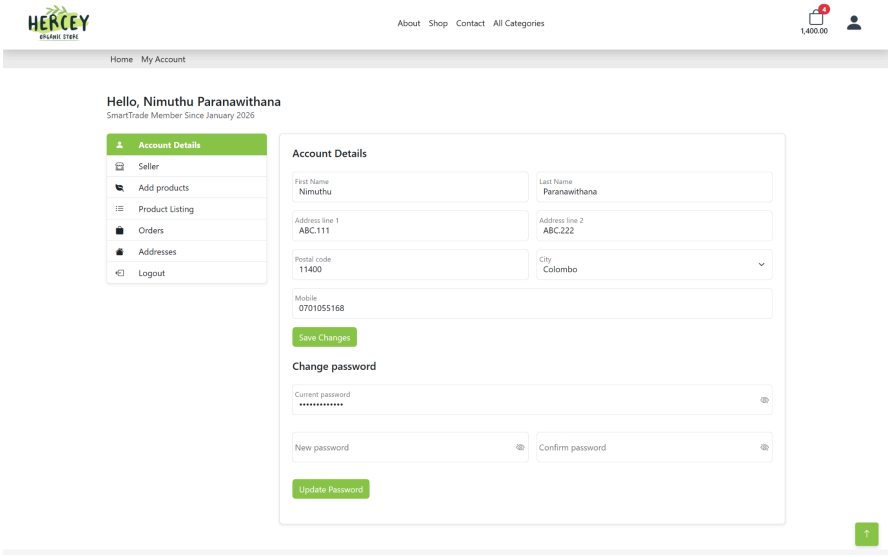
As mentioned earlier each product should have a state of approved before making it in to the store

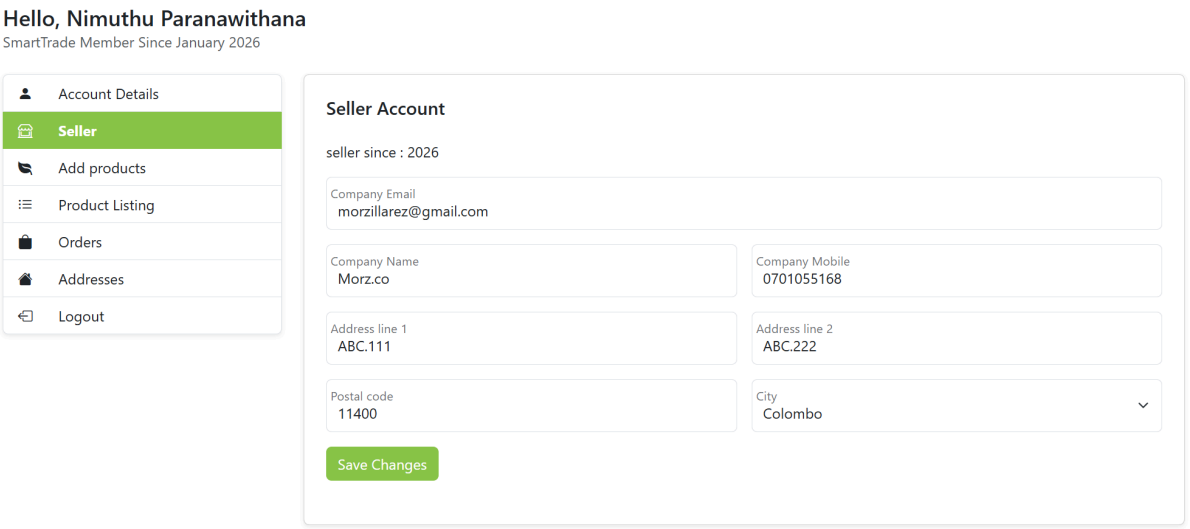


After home the next most prioritized section is the shop section because this is where the average user spends most of the time. There is a simple key work search for the user looking for a specific product and an advanced search with filters and range inputs for users that don't have a clear idea of what they are looking for. Both key words and the advanced filters work together to give the user a better experience.

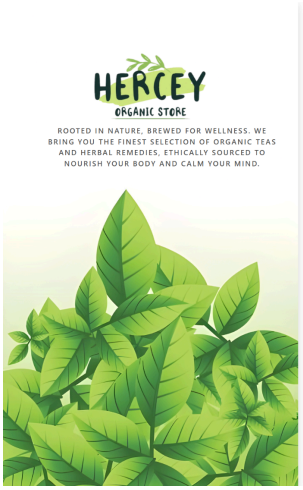


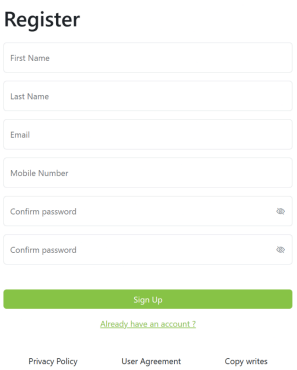
Next we have to profile section and the sub sections

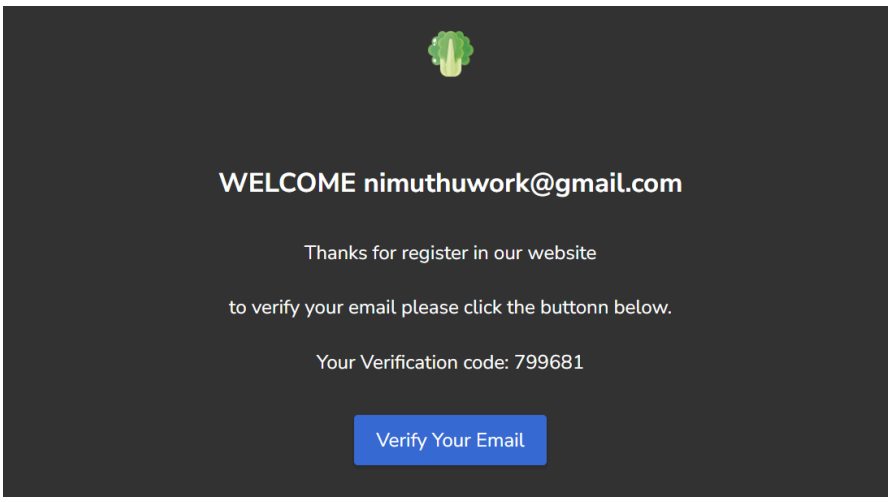


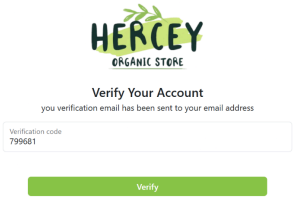


For a user to access this section the user must have an approved account verified with email.



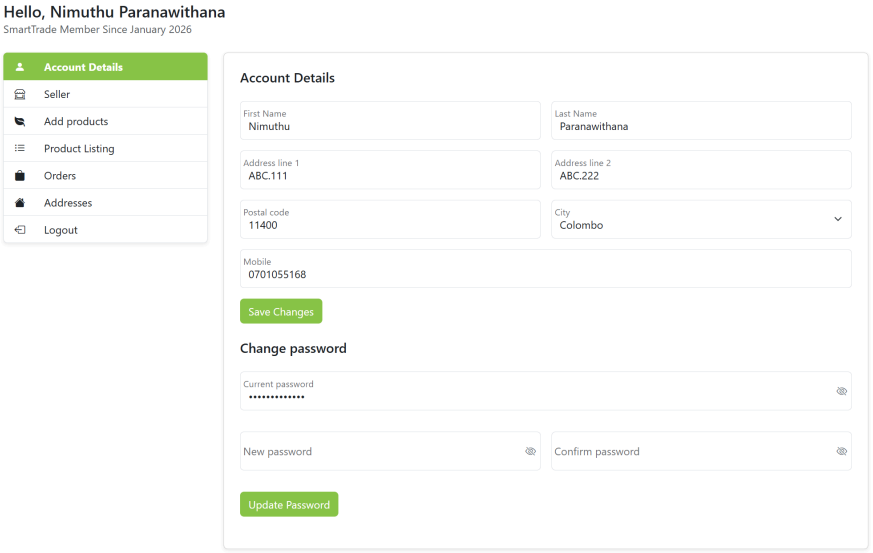




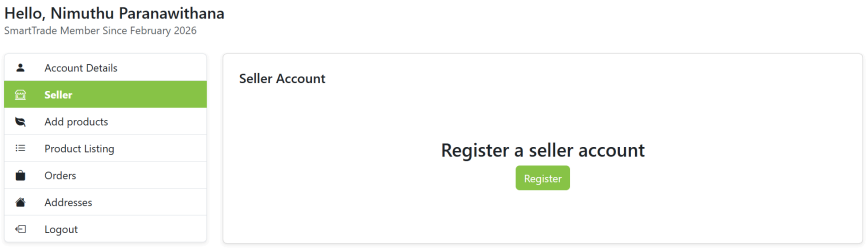




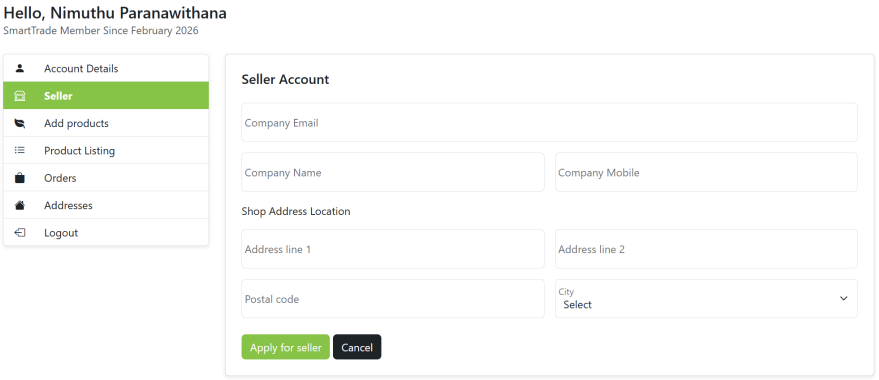
In this section a user can change their details and change their passwords. User can change any field other than the password without the password confirmation except for changing the password.



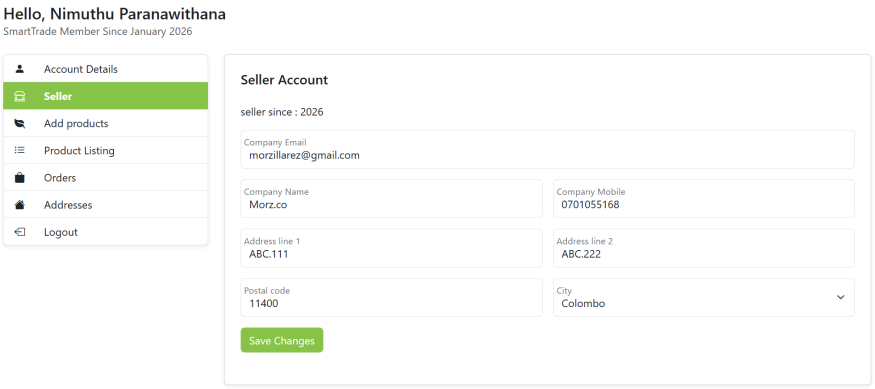
Next we have a seller account
A newly created user account cannot sell in the platform without being approved as a seller



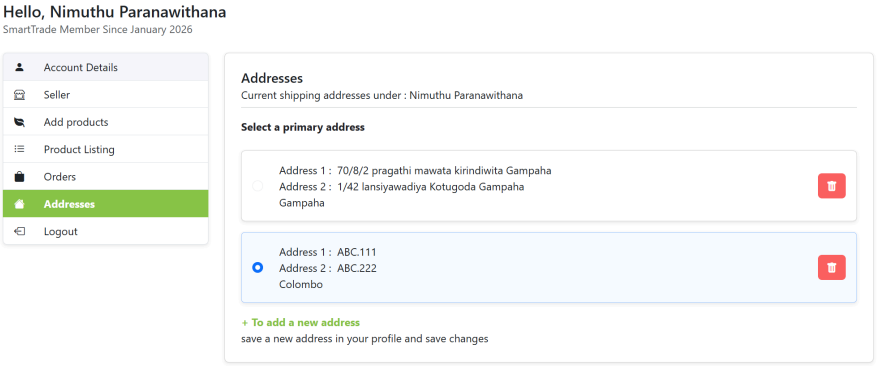
Apply for seller



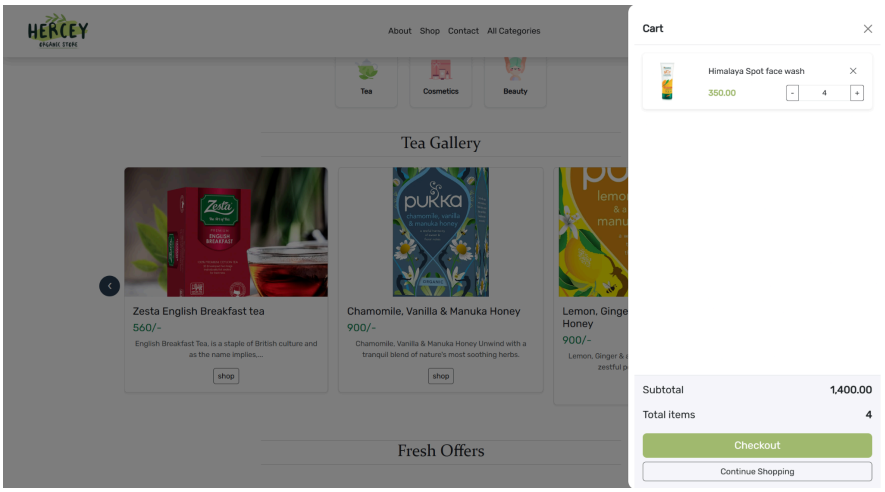
After approval



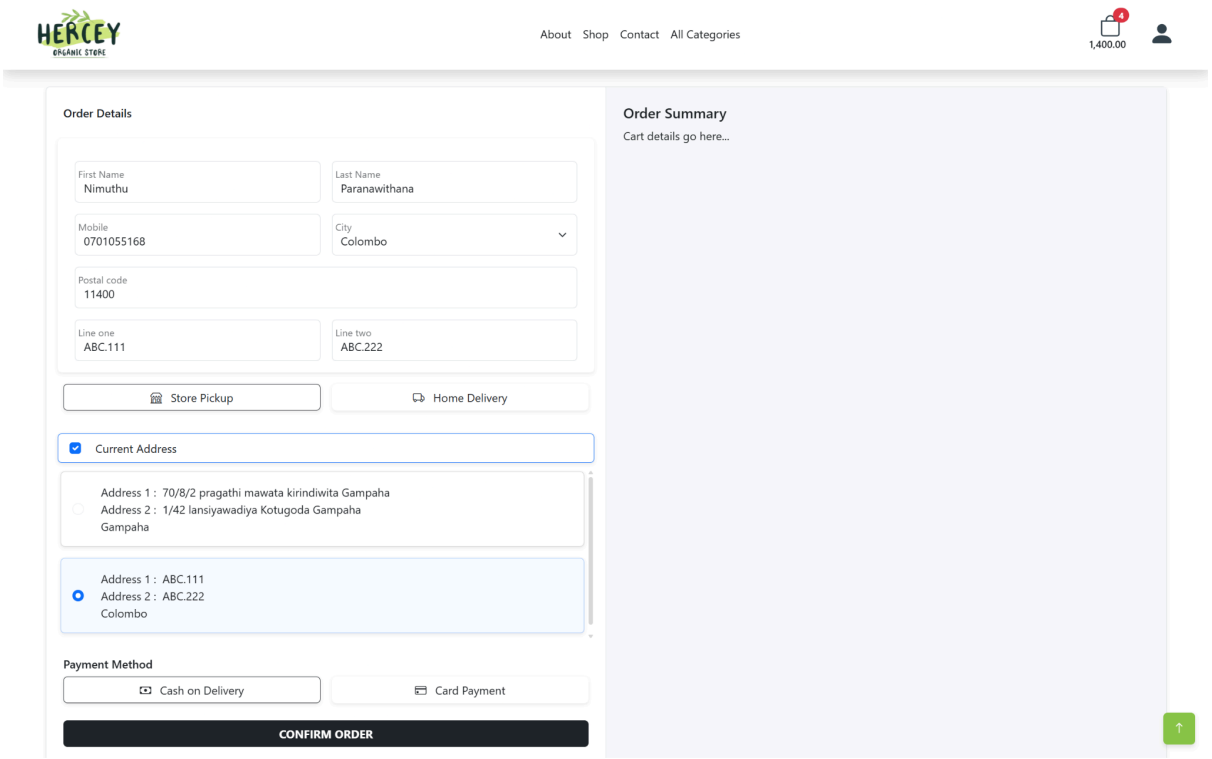
Then there are 2 section for user products listings and orders and then an address selection for the user to pick an address as the primary so that will load in every time the user enter check out



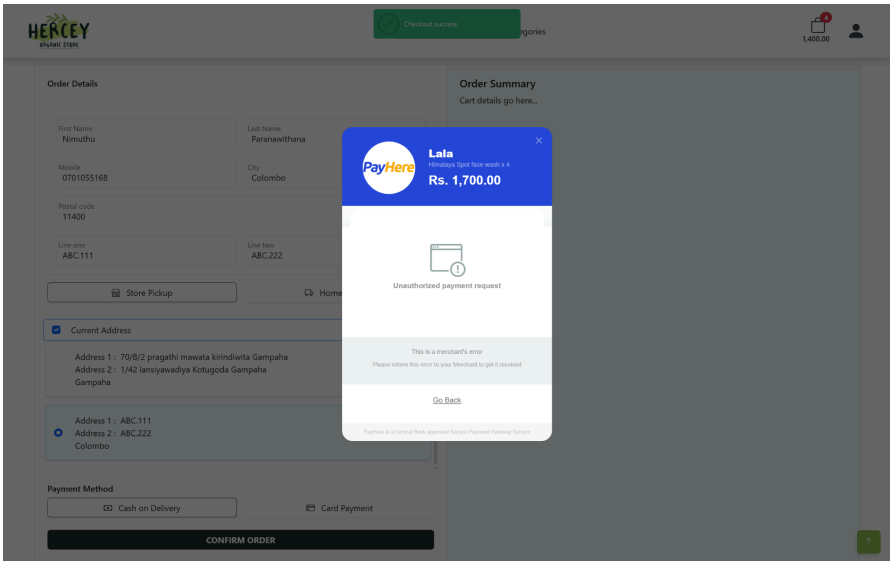
We have a side cart section where the user can edit cart items and remove items from the cart. If the user is satisfied with the items in the cart they can proceed to the checkout section from here



Upon check out the user will land on the check out section



Here they can change delivery details, payment options and the address. And on the left side the cart items are supposed to be there. When the checkout is pressed it will proceed to payHere payment interface



Where the user can enter the payment information and checkout.

5.2. External Interface Requirements

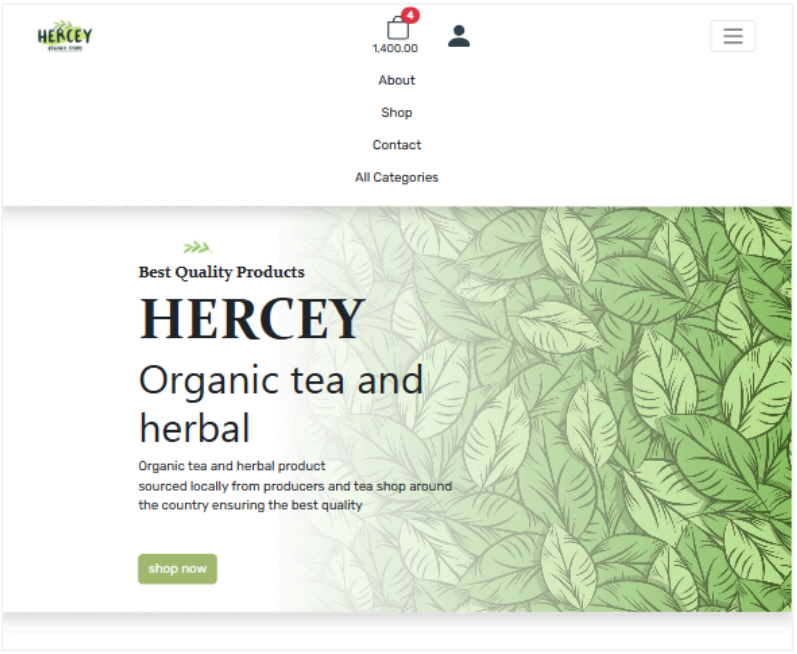
5.2.1. User Interfaces

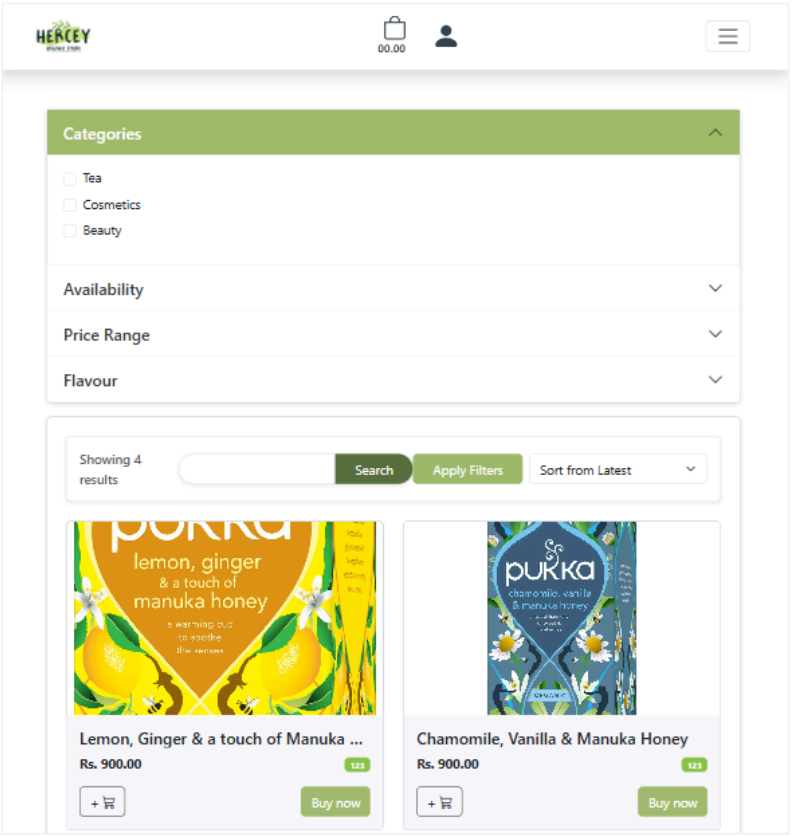
The web application is designed inspired by nature and herbal leaves.

There are conveniently placed error displays and error alerts to notify the user with errors, events, etc.

The web application is made to be responsive but the user's device should have at least a display that's 6.1 and 6.5 inches or 720x1280 in pixels. Or anything above

If the application is run on a small device, the user interfaces shift to compensate for the lack of width.





The side cart is made to meet all the requirements of a full cart page, it shows the product image and all the controls needed in a cart page.

Each item card has the quantity left, a few lines from the description, add to cart icon, primary product image, price and a buy now button.

The product details page includes up to 4 high quality images and the user can see each of them in a large view as well. It also includes all

the details about the product and navigation options to buy the product directly or add it to the cart.

5.2.2. Server Hardware

Client Side

This web application can run on almost any device that can run a standard HTML-5 web application. This includes desktops, hand held devices, tablets. A minimum of 4GB ram is recommended.

Server Side

The web application itself will be hosted on a server that can run tomcat server or any other java container software.

This system requires an uninterrupted TCP/IP network connection both in client side and server side to communicate through HTTP/HTTPS

5.2.3. Software Interfaces

Hercey uses a MySql database for Data storage. This application connects to the database using the JDBC connector. As for now the database is locally hosted and can be hosted elsewhere later when the app is launched.

Since the application is built with java it can run on any operating system that can run JVM (Java Virtual Machine).

This application uses an embedded tomcat server using Coyote (HTTP Connector) as the server and Catalina as the container software.

The server is built with a Java maven built tool to automatically retrieve libraries and plugins.

Here are the libraries, plugins thats being used buy this web application

- `Apache.tomcat.embed`
    - Like mentioned earlier tom cat is the server used to run this web application.
- `Tomcat-embed-jasper`
    - Jasper provides the JSP engine for rendering jsp pages in this web application
- `Jakarta.servlet-api`
    - The servlet api is for the back end processes happening in the controller since all controllers are servlets.
- `Hibernate`
    - Hibernate is a Java-based framework that maps object-oriented domain models to relational databases. This allows us to make an object oriented database.

- `Mysql-connector-j`
    - This is what connects the web application to the remote hosted database or local database depending on the development stage.
- `Jersey`
    - Jersey is what adds support to the server for REST API callings.
- `Gson`
    - GSON is used to convert json object and map json objects in to DTO objects in this application
- `Jakarta.mail`
    - Jakarta mail is what allows the application to send verification emails and to connect to the mail server which is mailtrap in this
- `Rocketbase.mail`
    - Rocketbase.mail is a email-template-builder that helps us developers build email templates to send dynamic bulk emails

6. Functional Requirements

6.1. Users should be able to login with their user name \ mobile number or email and the password

6.2. Users should be able to log in or sign up with google

6.3. The user should be able to change their password using email verification or mobile verification.

6.4. Users should be able to create an account with essential details only

6.4.1. User shouldn't have to spend time entering address data and country information unless the user is buying a product

6.5. Users should be able to verify account via email

6.6. Users should be able to view the web application as a guest user

6.7. Guest users should be able to add items to the cart

6.8. When a guest user login or creates an account the guest user cart should merge with the new account.

6.9. If the user is not registered, shall not be allowed to access account and checkout pages.

6.10. User should have quick access to end points mentioned below upon arriving at the home page

6.10.1.1.1. About

6.10.1.1.2. Chat support

6.10.1.1.3. B2B

6.10.1.1.4. Store

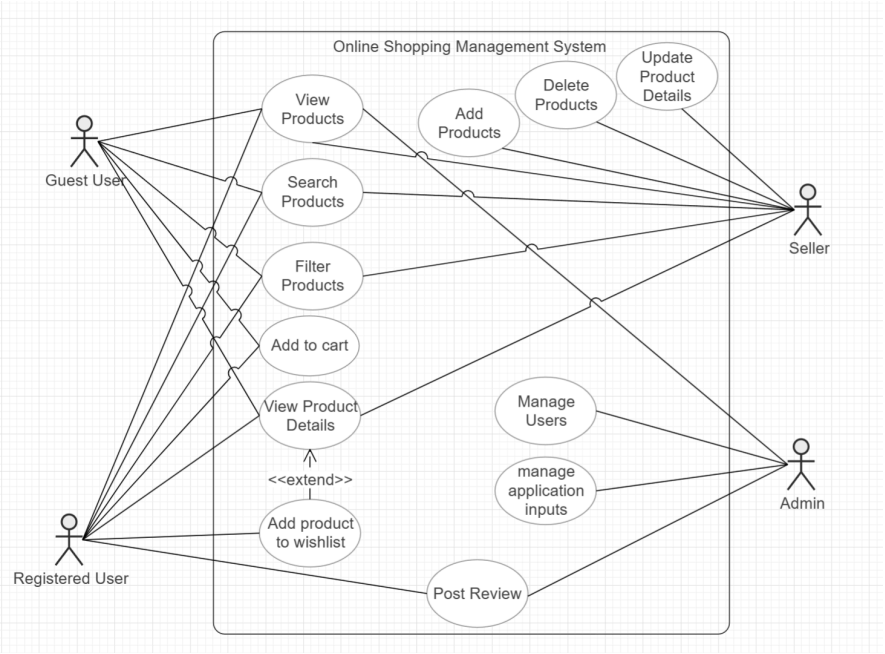
6.10.1.1.5. Fresh deals

6.10.1.1.6. All categories

6.10.1.1.7. Tea stage (Gallery)

6.11. Users should be able to delete their account

7. Behavior Requirements



A guest user can view, search products, add products to cart. A guest cannot access a wish list. A guests products are added to a session cart and when the user create or log in to an account the DB cart is moved to the DB cart

A Registered user can view, search products, add products to cart and add items to wish list. A registered user's products that are being added to the cart will be saved directly in the cart database.

A seller can add products, edit products, view product listing, view orders list

An admin can manage users including customers and sellers. Admins can manage the application, for example admins can add new categories to the application.

8. Performance Requirements

The user must have a stable internet connection with a download speed of at least 100kbps and a download speed of at least 50kbps to run this website.
In standards of 2026, probably even a smart watch can run this web application.

The user must have a suitable PC or a suitable mobile or a device that can browse the web. With at least 400Mb headroom of RAM since the full website on itself takes close to 390Mb but with an optimized web browser it can still work on 100Mb headroom but we cannot guarantee the performance.

The application needs to be hosted in a server that has an uptime as much as 100%. The server or the partition should (virtual machine) should at least have (Assuming there will be 10 000 users connecting at the same time).

CPU: 32+ Cores like a Dual AMD EPYC or Intel Xeon Gold to handle thousands of Java threads at the same time.

RAM: 128 GB – 256 GB to accommodate JVM Heap, Database caching, and OS overhead.
Storage 1 TB+ NVMe SSDs with high read write speeds for high I/O database transactions.

Network 10 Gbps bandwidth to handle API calls and response and static content deliveries .

The server needs to be seamlessly integrated with the client side and

9. Safety and Security Requirements

The web application always uses email verification to authenticate the user in a situation where the password is forgotten.
And all authentication processes go under HTTPS protocol. All sensitive user data are delivered by POST body and data like passwords are encrypted before sending and storing.
Users can review the service and the website gets updates accordingly to ensure data safety, performing multiple penetration tests throughout the year.
The website requires the user to accept return and purchase policies before the first transaction.

10.